Project Title: ***Optimal hospital selection for EMS transport of COVID-19 patients***
Team Number (Team Members): ***176 (Prateek Gowda, Ruchita Kothari, Alex Lu, Nicole Pagane, Kate Schole, Utkarsh Sharma)***
Selected Prompt/Need: ***Hospitals need a way to increase their capacity to physically accomodate an increased number of COVID-19 patients***

1. Problem Overview / Relevant Clinical Background

The COVID-19 pandemic has begun to overwhelm our healthcare system over the course of just a few months. Current predictions claim that the United States will exceed ventilator and ICU bed capacity as early as the second week of April 2020 [1]. Simultaneously, Emergency Medical Services (EMS) have seen a record-breaking number of calls during this pandemic. Within one day in late March NYC EMS received more than 7000 calls, which is more than were seen on September 11th, 2001 [2].

This increasing number of COVID-19 patients is projected to result in striking ICU bed and ventilator shortages within hospitals [3]. Not only are these resources in short supply, but their allocation is also drastically uneven between hospitals in urban settings such as New York City **[Figure 1]**. Thus, to provide the most efficient care delivery possible, patient distribution across a hospital system should mirror the distribution of hospital supplies.

Presently, however, EMS does not choose its destination based on hospital capacity, but rather prioritizes hospital proximity and patient preference [4]. Additionally, when Emergency Departments (EDs) and Intensive Care Units (ICUs) are overwhelmed, EMS encounters longer wait times and may be asked to transfer patients to an alternative center with more resources [5]. This inefficiency impairs patient care by straining ED and EMS staff, and this burden will be significantly exacerbated by the current COVID-19 pandemic. **Therefore, in urban environments with multiple hospitals, there is a need for EMS first responders to quickly select the most appropriate hospital for each suspected COVID-19 patient, based on the hospital's COVID-19 care capacity.**

2. Proposed Solution

We propose a mobile application, Ambumetrix, for EMS use that facilitates optimal selection of a hospital for a given patient based on comprehensive hospital capacity metrics rather than simply on minimal distance. This promotes balanced utilization of all hospitals within a given region, reducing inefficiencies secondary to premature hospital saturation. Ambumetrix works as follows **[Figure 2]**:

1. **Input** — EMS will input features such as suspected COVID status, as well as signs that highlight respiratory distress including respiratory rate, color changes around mouth/lips, chest retractions, body position, etc. [6].
2. **Hospital ranking algorithm** — The algorithm combines EMS inputs with characteristics of local hospitals, such as availability of ventilators (prioritized if respiratory distress is observed), beds, testing kits, ED wait times, and proximity, to rank hospitals by capacity for delivering care.

3. **Hospital contact** —The ranked hospital list segues into outgoing calls for EMS to directly communicate with ED charge nurses.

3. Use Case

We intend for EMS responding to patients suspected of being COVID-19 positive to be the primary users of our mobile application. There are few barriers to use with minimal workflow disruption. Patient parameters input into the application do not significantly differ from those that would otherwise be gathered in standard examination. Communication from EMS crews to hospital EDs parallels workflows in the standard of care.

4. Evidence for Functionality/Efficacy.

A 2005 retrospective case study correlates hospital bed capacity tracking with decreased EMS ambulance diversion time using an Internet-based monitoring system in Western Australia [7]. We extend this with additional capacity parameters and a decision-based algorithm.

To assess the impact of Ambumetrix, we compare it to the standard model of EMS response by simulating a city with nine hospitals of varying capacities; see Appendix for model specifications **[Table 1]**. While the standard model routes patients to the nearest hospital, Ambumetrix applies capacity-informed decision criteria to choose the most appropriate hospital. Numerical results show that in addition to Ambumetrix evenly distributing the city's patients across all nine hospitals **[Figures 3 and 4]**, it significantly lowers wait times for the ED and for critically-ill patients in need of mechanical ventilation **[Figures 5 and 6]**. Further, this wait time efficiency allows for increased number of patients to be treated **[Figure 7]**. While the standard approach can only treat about 3 in 4 patients in a given day (74.1%) with our simulated parameters, Ambumetrix is able to treat nearly all of the city's patients (97.85%).

While we observe dramatic failure in both Ambumetrix-assisted and standard EMS response when cities reach maximum capacity, we expect Ambumetrix to be superior as pop-up hospitals become more prevalent. Ambumetrix will reroute patients to these centers to maximize city utilization of limited resources that are in flux. Evident from our results, a city divided cannot stand, while the unification of hospitals through an optimally shared-distribution of patient load seems mightily difficult to topple.

In addition to model verification, we have received a letter of support from the Systems Director of EMS at Mount Sinai, which confirms the potential for efficacy of Ambumetrix (attached in final submission).

5. Further Design / Testing Work Required

Further development goals involve tuning the hospital ranking algorithm, developing input streams for hospital capacity metrics, and live deployment.

1. **Tuning hospital ranking algorithm** — Our current algorithm maps EMS inputs and hospital characteristics to a predicted optimality metric for each hospital. Input features

were informed by knowledge of policies on response to suspected COVID cases and common COVID presentations. Although this performs effectively in numerical simulations, tuning and validation against real-world hospital / EMS data would improve feature selection and algorithm performance.

2. **Developing input streams for hospital capacity metrics** — With permission from each hospital, necessary capacity metrics can be extracted from hospital EMR systems [8, 9].
3. **Live deployment** — The ranking algorithm and hospital capacity data must be integrated as a backend to the existing application front end.

According to FDA guidelines on mobile medical applications, Ambumetrix is not subject to FDA regulation [10]. Protected Health Information (PHI) is neither shared nor stored, allowing HIPAA compliance. We will partner with each state's Department of Health to guide interoperability procedures and facilitate implementation.

6. Implementation Plan

For rapid deployment and impact, we will conduct a beta development period within the next month, followed by a week-long multi-center trial period in New York, and then implement widely in all of NYC through partnership with the NY State Department of Health. Ambumetrix can then be translated to the many other cities impacted by COVID-19.

1. **Pilot** — A key component of our technology is access to numbers of hospital beds, ventilators, COVID-19 tests, and wait times through EMR integration. We have confirmed that this is possible at JHU through interviewing Peter Greene, Chief Medical Information Officer of Hopkins. Therefore, we plan to pilot our prototype at Johns Hopkins Hospital and integrate feedback into a beta product by the end of April 2020.
2. **Beta** — NYC is an area that will benefit from optimized ambulance transport for COVID-19 patients. All major NYC hospital systems also use Epic, allowing for smooth database transition (confirmed by an Epic informaticist at Columbia University Irving Medical Center).

In addition, this technology falls within use of the ET3 payment model - a federal healthcare plan implemented in February 2020 that incentivizes all major NYC hospitals to coordinate care and match patients to facility by patient acuity [11]. Thus, there exist both cost-reduction and resource shortage motivations for Ambumetrix adoption among NYC EMS providers.

7. Resources Needed for Completion

We are in contact with physicians and other medical staff through Hopkins who can provide feedback on the best way to implement our application in the ED. Additionally, we will need to develop partnerships with EMS systems in NYC. We will also need government and medical sponsors for financial support, as well as backend software developers for database management and implementation of the application. We will continue to interface with first responders, emergency medicine physicians, and nurses for guidance. We will also need the support of EMR database managers to determine the best way to receive inputs from hospitals on their current supply capacity and ED wait times.

Appendix:

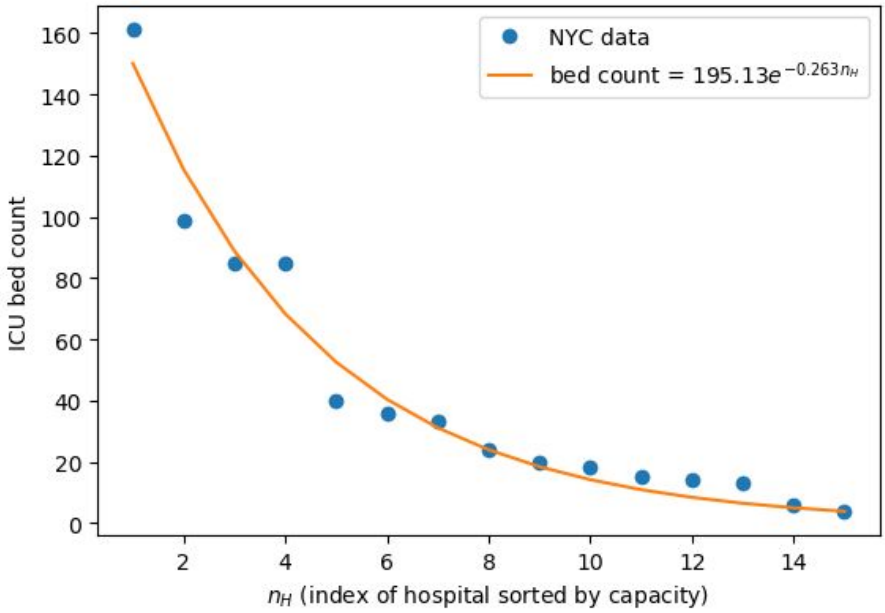

Figure 1: Distribution (mean = 45.54, standard deviation = 42.95) of ICU bed count in Manhattan, NY. Evident by the exponential fit, there is drastic uneven capacity allocation across city hospitals.

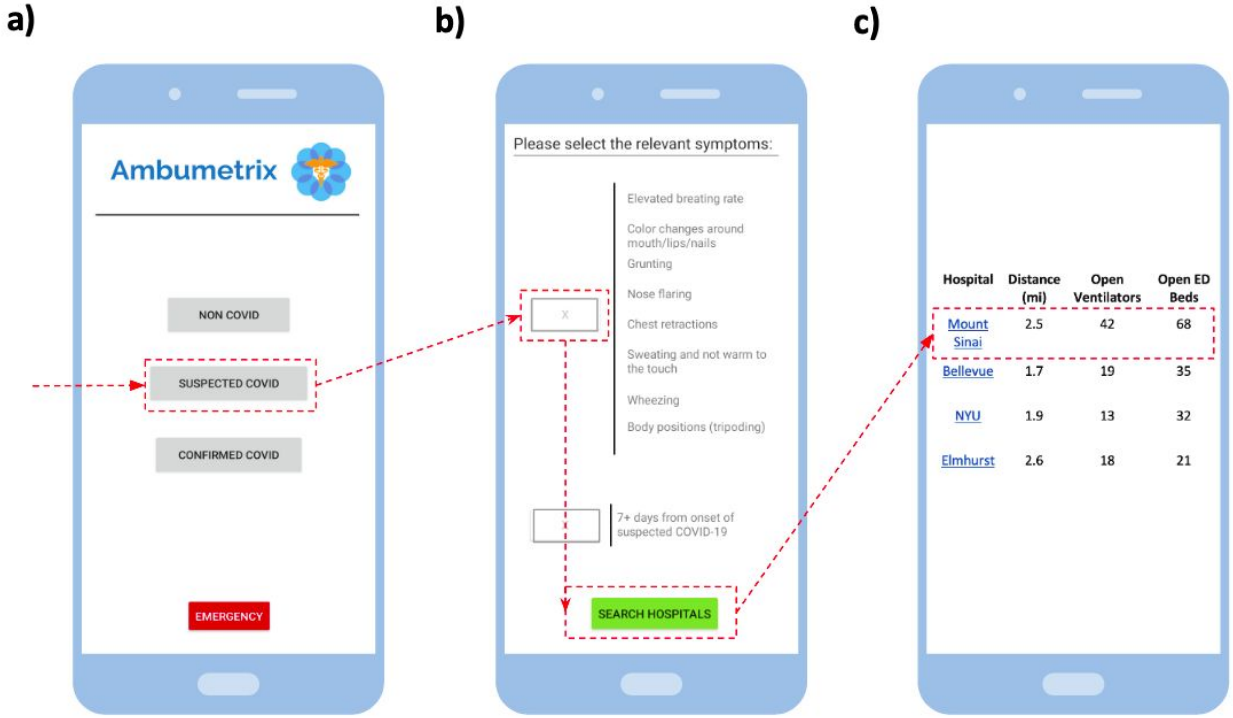

Figure 2: Workflow of Ambumetrix Android App GUI implementation in hypothetical NYC EMS response. **a)** The first screen requests the EMS user to classify their patient by COVID status. It is important to note that if the patient is acute and unstable, the EMS user can click Emergency, in which case hospitals will be listed in order of proximity. If the patient is non-COVID and stable, a hospital list will be shown that can be sorted by various parameters. If the EMS user selects suspected or confirmed COVID, the next panel **b)** requests the user to identify whether the patient shows symptoms of respiratory distress or whether the patient has had COVID-like symptoms for more than 7 days. Both of these parameters indicate possible need for a ventilator [12]. **c)** Once symptoms are submitted, the app generates a ranked list of hospitals based on the necessary parameters (ventilators, beds, proximity, wait time). Note that the number of tests is only used as a parameter if the patient is suspected-COVID, and has not already been confirmed COVID positive.

Model Specifications for Simulation:

- Total number of hospitals: 9
- Total number of patients : 4000
- ED size: [20-60] beds per hospital (randomly generated)
- ICU size: [7-20] beds/ventilators per hospital (scaled to ⅓ of respective ED size)
- Hospital locations: randomly distributed on square grid
- Patient call locations: randomly distributed on square grid
- Patient health: [0-1] (randomly generated) akin to symptom classification. Higher values correlates with better health
- Patient influx: hospitals perform triage to assign patients to necessary departments if there is availability
- Patient efflux: hospitals discharge from both the ED and ICU with given probabilities inspired from COVID-19 patient outcomes

*all random numbers were sampled from a uniform distribution

Table 1: For more information on model implementation and Android app design, see our GitHub repo: https://github.com/npagane/CBID.

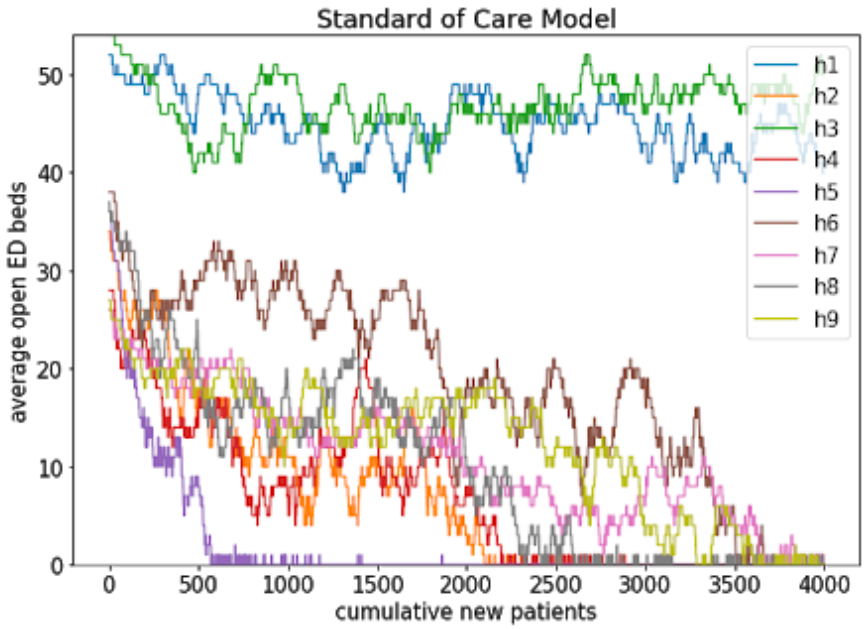

Figure 3: Trajectories of available ED beds across the nine simulated hospitals for a cumulative patient count of 4000 across the entire city using the standard EMS response. Smaller hospitals quickly become overwhelmed by patients, while larger hospitals are underutilized.

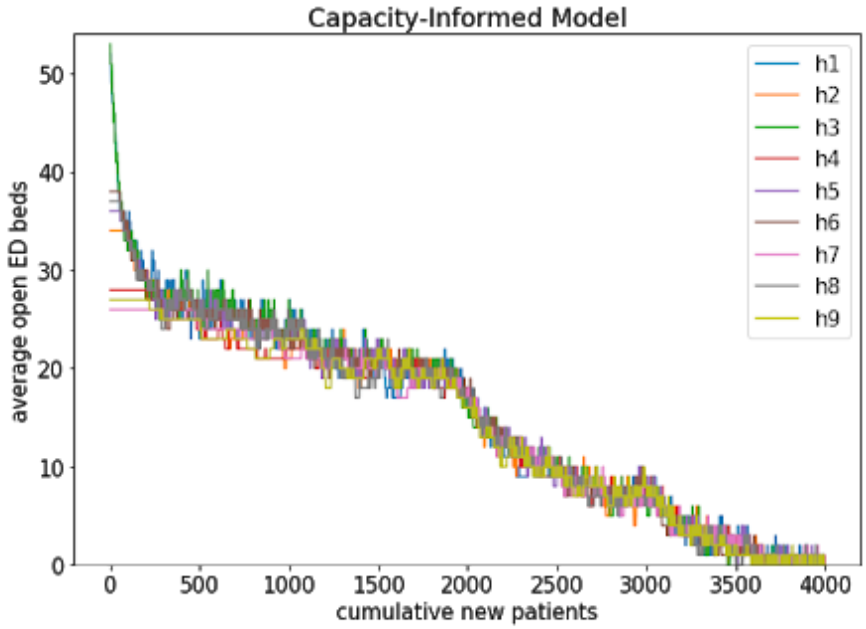

Figure 4: Trajectories of available ED beds across the nine simulated hospitals for a cumulative patient count of 4000 across the entire city using the Ambumetrix decision criterion described in the main text. The patient load is distributed evenly across the city hospitals.

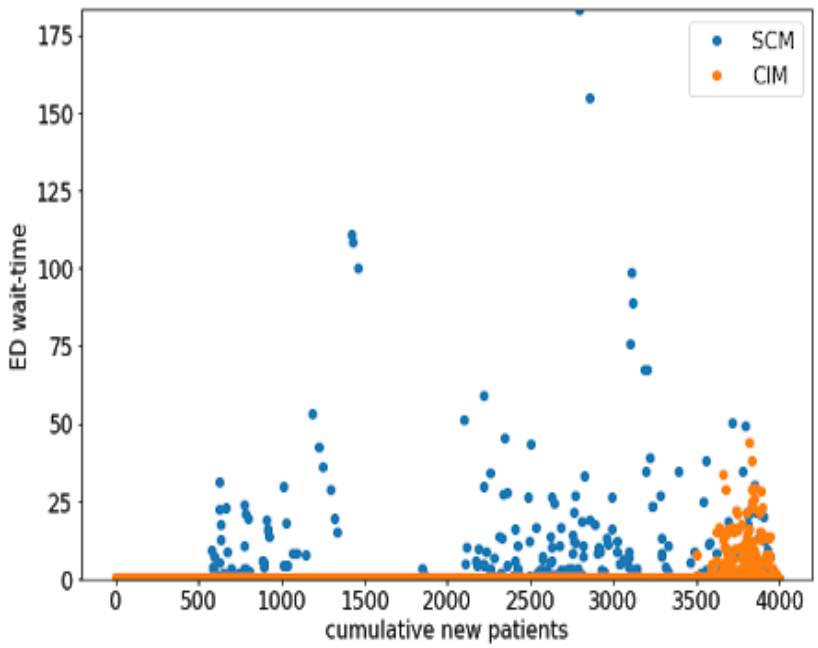

Figure 5: Average ED wait times across the city for the standard care EMS response (SCM, blue) and Ambumetrix (CIM, orange). We approximate 4000 patients to be 20 hours such that we get 200 patient calls across the city per hour. Thus, 1 ED wait-time corresponds to 3 minutes, and the average wait-time is 75 minutes. These relative values are valid for cross-model comparison, but are meaningless in an absolute interpretation without setting more real-world parameters.

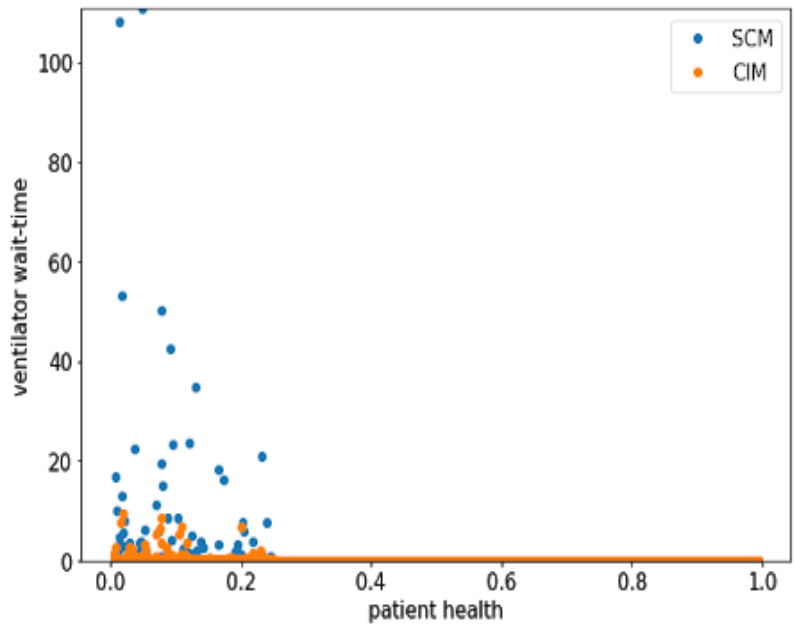

Figure 6: Average ventilator wait times across the city for standard EMS response (SCM, blue) and Ambumetrix (CIM, orange). See the above figure for proper interpretation of wait times. Only patients with health <0.25 were eligible to be admitted to the ICU and put on a ventilator.

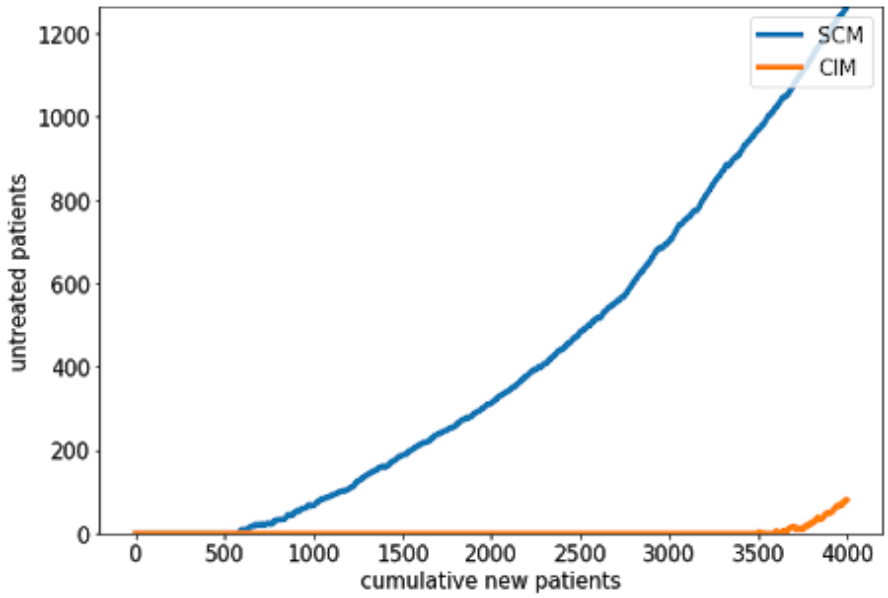

Figure 7: City-wide number of untreated patients (those remaining in triage without being seen by an ED physician) throughout the simulation for the standard care EMS response (SCM, blue) and Ambumetrix (CIM, orange). The final numbers reached on each curve is the total number of patients remaining untreated by the end of simulation. Finding the difference between these untreated values with the total number of patients yields 3,914 (Ambumetrix) and 2,964 (standard care) total patients taken care of by both models. Hence, Ambumetrix cares for 97.85% of the city's patients, while the standard care EMS response can only treat 74.1%.

Citations

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[12] Wang, Dawei et al. "Clinical Characteristics of 138 Hospitalized Patients With 2019 Novel Coronavirus-Infected Pneumonia in Wuhan, China." JAMA vol. 323,11 (2020): 1061-9.